

# CONTENTS

## TECHNOLOGY

- 22 INTERVIEW: "We Refer To The 28-Nanometre Node As The 'Goldilocks' Node" — Sharan Srinivas J. And Shashwath T.R., Co-Founders, Mindgrove Technologies

## MOVES

- 30 TECHNOLOGY: What On Earth Is A Software Defined Car?
- 34 INTERVIEW: "We Have Maintained A Net NPA Rate Below 1%!" — Sameer Aggarwal, Founder And CEO, RevFin
- 36 EVERYONE: Electric 2-Wheelers Priced Under ₹100,000
- 42 BUSINESS: With Electric Vehicles, The Vehicle Service Industry Will Undergo The Biggest Disruption



## STRATEGY

- 54 MY STORY: "Challenges Are The Biggest 'Change And Improve' Points In Life" — Vineet Sahni, FIEM Industries
- 60 TECH: Application Of Artificial Intelligence In The Manufacturing Sector
- 64 POLICY: India's Semiconductor Manufacturing Ambition Needs A Decadal Plan
- 68 GROWTH: Six Steps To Building Better Business Assets



## DESIGN

- 72 INTERVIEW: "Our Approach Involves Not Only Designing The System But Also Creating Test Setups For Various Scenarios" — Rijin John And Dr Arun Ashok, Co-Founders, Silizium Circuits
- 74 SECURITY: Safeguarding Chip Security: Challenges And Countermeasures
- 80 CIRCUIT: Interesting Reference Designs Of Compressors

## DO-IT-YOURSELF

- 82 CIRCUIT IDEA: Musical Rain Alarm
- 85 PROJECT: Smart Home Automation System Using ESP8266 NodeMCU
- 90 CIRCUIT IDEA: Light-Activated Robot
- 92 PROJECT: Design Your Own Low Cost RISC-V System
- 95 PROJECT: Counting Small Items Using Edge Impulse
- 98 CIRCUIT IDEA: An Electronics Kaleidoscope

## REGULARS

- 7 Research & Innovation Updates
- 11 Truly Innovative Electronics
- 14 New Products
- 28 Electroons 2.0
- 44 Dashboard: Key Industry Trends That Can Shape The Future Of Your Business
- 46 Startups & Innovators
- 50 Industry News: Industry Updates

**EDITOR** : Rahul Chopra

**EDITORIAL CORRESPONDENCE** : Phone: +91-9811155335  
E-mail: editop@efy.in

**SUBSCRIPTIONS & MISSING ISSUES** : Phone: +91-9811155335  
E-mail: support@efy.in

**BACK ISSUES, BOOKS, KITS, COMPONENTS, etc** : Phone: +91-9811155335  
E-mail: support@efy.in

**NEWSSTAND DISTRIBUTION** : Phone: +91-9811155335  
E-mail: efycirc@efy.in

**ADVERTISEMENTS**  
NEW DELHI : Phone: +91-9811155335  
(HEAD OFFICE) E-mail: efyenq@efy.in

MUMBAI : E-mail: rmowest@efy.in

BENGALURU : E-mail: rmosouth@efy.in

CHINA : Worldwide Focus Media  
E-mail: china@efy.in

GERMANY : PMS Plantenberg Media Service GmbH  
E-mail: germany@efy.in

JAPAN : Tandem Inc.  
E-mail: japan@efy.in

TAIWAN : J.K. Media  
E-mail: taiwan@efy.in

UNITED KINGDOM : ASA Media  
E-mail: uk@efy.in

UNITED STATES : E & Tech Media  
E-mail: usa@efy.in

Printed, published and owned by Ramesh Chopra. Printed at M.P. Printers, B-220, Phase-2, NOIDA-201305, Uttar Pradesh, on the first day of each month and published from D-87/1, Okhla Industrial Area, Phase-1, New Delhi 110020. Copyright 2023. All rights reserved throughout the world. Reproduction of any material from this magazine in any manner without the written permission of the publisher is prohibited. Although every effort is made to ensure accuracy, no responsibility whatsoever is taken for any loss due to publishing errors. EFY will not be responsible for any wrong claims made by an advertiser. Disputes, if any, will be settled in a New Delhi court only.

## SUBSCRIPTION RATES

EFY = Electronics For You

| PRINT MAGAZINE                   | 1 YEAR (12 copies each) | 3 YEARS (36 copies each) | 5 YEARS (60 copies each) |
|----------------------------------|-------------------------|--------------------------|--------------------------|
| WITHIN INDIA (IN RUPEES)         |                         |                          |                          |
| EFY (Rs 100/copy)                | 840                     | 2150                     | 3000                     |
| SAARC COUNTRIES (IN US\$ BY AIR) |                         |                          |                          |
| EFY                              | 50                      | 135                      | NA                       |
| OTHER COUNTRIES (IN US\$ BY AIR) |                         |                          |                          |
| EFY                              | 100                     | 270                      | NA                       |

Courier delivery within India @ Rs 50 per copy

e-magazine subscriptions within India are available at half the rates mentioned here.

Overseas rates for each e-magazine in US\$:

1 year: \$12; 3 years: \$33; 5 years: \$50 only

Please send payments only in favour of **EFY Enterprises Pvt Ltd** through demand draft/cheque/money order to D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020, or online at: <http://subscribe.efyindia.com> or scan the QR code.

